

QUANTUM SENSORS: The New Era

Quantum sensors—masters of precision, exist in multiple states simultaneously using superposition and entanglement, guided by laser light for meticulous control.
They are now finding wide-ranging applications



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In plant physiology and horticulture, light is pivotal for plant growth. However, not all light is equal in its impact on plant photosynthesis and health. Photosynthetically active radiation (PAR), the 400 to 700 nanometre light wavelength range, is crucial for photosynthesis.

PAR quantum sensors, using advanced photodiode technology, accurately measure PAR intensity. They utilise advanced photodiode technology that can accurately detect and measure the number of photons within the PAR range. Equipped with a light-sensitive photodiode, signal amplifier, and digital interface, these sensors provide precise measurements of light energy for optimal plant cultivation and greenhouse management. By understanding the PAR levels in different growth environments, researchers can optimise plant cultivation techniques, enhance crop yields, and study the effects of light on various plant species.

PAR quantum sensors offer several advantages over traditional light meters.

First, they are specifically calibrated to measure light within the PAR range, ensuring accurate readings for photosynthetic activity. Second, these sensors are often compact and portable, allowing for easy deployment and data collection in various settings. Many PAR sensors also offer real-time monitoring

capabilities, enabling researchers to track fluctuations in light intensity over time.

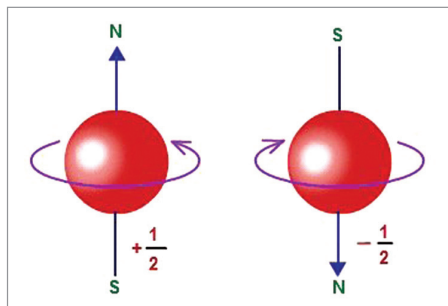
The data collected by PAR quantum sensors can be used to calculate important parameters, such as photosynthetic photon flux density (PPFD), which quantifies the number of photons reaching a specific area per unit time. PPFD is a critical metric for determining light requirements, optimising lighting systems, and understanding plant responses to different light conditions.

Working

Quantum sensors use quantum mechanics principles for precise measurements. The working principle of quantum sensors involves several key concepts and techniques.

One important concept in quantum sensing is superposition. According to quantum mechanics, particles can exist in multiple states simultaneously. In the case of quantum sensors, this means that the sensor can be in a superposition of different states, each corresponding to a different value of the measured quantity. By manipulating and measuring the superposition state, quantum sensors can obtain precise information about the physical parameter being measured.

Another crucial concept is entanglement. Entanglement occurs when two or more particles become correlated in such a way that the state of one particle is directly linked to the state of the others, regardless of the distance between them.



Two spin natures of the electron, basically positive and negative spin half